07E — Deep-Dive Feature Audit: Sect Governance & Internal Ecosystem (Part 5 of 5 — Grand Finale)

Master Holistic Overview of the Entire Living Sect Ecosystem: `SIM.retire` & `graveyard` Corruption Bugs, Save/Load Desync, Closed-Loop Macro Fiscals, Officer-Mediated Automation & Master `P0` $\to$ `P3` Roadmap

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` in Google Drive `Js2 new`)
- **Target Audience**: Autonomous AI Developers (**GPT** & **Gemini**)
- **Role of Document `07E`**: **Master Holistic Synthesis** concluding the 5-part Sect Governance & Internal Ecosystem Audit (`07A–07E`), inspecting the master simulation spine (`js/sim.js`) where all 25 offices, 8 Sacred Peaks, buildings, politics, courts, lineages, and fiscal ledgers converge.
- **Immutable Project Rules**:
  1. **NEVER modify `js/human_core.js`** (`HC.FINGERPRINT === '155:2f9c9dd7'`).
  2. **Preserve exact 40 `<script>` tags** and load order in `index.html` / `test/load.js`.
  3. **100% Emergent NPC Causality & Post-Load Determinism**.

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1. `[SECT-SYS-01]` Critical Lifecycle & Conservation Bugs in `SIM.retire` (`js/sim.js:347–385`): Traitors Donate Secret Scrolls, Spirit Stones & Pets Vanish, and Dangling IDs Corrupt Peaks/Leylines/Outposts!

Inspect `SIM.retire(W, n)` in `js/sim.js:347–385` line-by-line:

```javascript
for (const k of Object.keys(n.equip || {})) { if (n.equip[k])
W.sect.gear.push(n.equip[k]); } n.equip = {};
...
for (const id of ((n.inv && n.inv.techs) || [])) if (CULT.tech(id) &&
!W.sect.library.includes(id)) W.sect.library.push(id);
```

```
for (const id of ((n.inv && n.inv.recipes) || [])) if (ECO.recipeById[id] &&
!W.sect.knownRecipes.includes(id)) W.sect.knownRecipes.push(id);
...
if (i >= 0) W.npcs.splice(i, 1);
if (W.byId && W.byId[n.id] === n) delete W.byId[n.id];
```

Four Severe Bugs in `SIM.retire`:

1. **Exiled Rebels Donate Their Secret Hoarded Manuals & Recipes (`lines 364–365`)**:
   - A treacherous disciple (`n.exiled === true`, `loyalty < 20`) who refused to donate a rare `TECH_MANUAL` or `RECIPE_SCROLL` is expelled from the sect—and `SIM.retire` **hands their secret manual and recipe straight into `W.sect.library` and `W.sect.knownRecipes` for free**!
   - **Contextual Fix**: Only donate `inv.techs`/`inv.recipes` on **peaceful retirement (`!n.exiled`)**; when `n.exiled === true` (`violent betrayal/expulsion`), confiscate Sect-owned property if captured by `discipline`, or let fleeing defectors carry their personal scrolls/stones into a rival/splinter sect in `W.world.sects`!
2. **`100%` of Personal Spirit Stones (`n.inv.stones`) & Bonded Pets (`n.pets`) Vanish Into Thin Air**:
   - `SIM.retire` processes `equip`, `gear`, `pills`, `mats`, `techs`, and `recipes`—**completely forgetting `n.inv.stones` and `n.pets`**! Splicing `n` out of `W.npcs` permanently deletes all Spirit Stones in `n.inv.stones` (violating Closed-Loop conservation!) and deletes living `n.pets` instead of releasing them to `W.sect.reservePets` or taking them into exile!
3. **Dangling Ghost IDs Left Behind in `PEAKS`, `LEYLINE_GRID`, `OUTPOSTS` & `closedDoorSeclusion`**:
   - Lines `373–381` clean `deputyId`, `successorId`, and `resourceNodes`, **but NEVER clean**:
     - `W.sect.peaks.byId[*]` (`masterId`, `deputyId`, `memberIds`) $\to$ Peak Master points to `undefined`!
     - `W.sect.leylineGrid.anchors[*].guardNpcId` $\to$ Ghost guard!
     - `W.sect.outposts.byNodeId[*].garrisonNpcIds` $\to$ Ghost garrison!
     - `otherNpc.closedDoorSeclusion.protectorId` & `otherNpc.companionId`!

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2. [SECT-SYS-02] Double Save/Load & Schema Crash Bug on Graveyard Burial (`js/sim.js:488` vs `583` & `608`)!

Inspect `js/sim.js:488` (`SIM.dailyUpdate` 4-day corpse burial) alongside `js/sim.js:583` and `608` (`SIM.deserialize`):

```javascript
// Line 488 (SIM.dailyUpdate):
if (n.dead && W.day - n.deathDay > 4) {
  SIM.snapshotPerson(W, n, "เสียชีวิตและฝังในสุสานสำนัก");
  W.npcs.splice(i, 1);
  W.graveyard.push(n.id); // <-- PUSHES A STRING ID ('n2_415'), AND DOES NOT DELETE W.byId[n.id]!!
}

// Line 583 (SIM.deserialize validator):
const inGrave = Array.isArray(s.graveyard) && s.graveyard.some(g2 => g2 && g2.id === s.sect.masterId);
```

1. **Type Mismatch Crash in `SIM.deserialize` (`line 583`)**:
   - `W.graveyard` stores **plain string IDs (`'n2_415'`)**, yet line `583` checks **`g2.id === s.sect.masterId`** (`'n2_415'.id` is `undefined`!)! Fix line `583` to `(typeof g2 === 'string' ? g2 === s.sect.masterId : g2 && g2.id === s.sect.masterId)`!
2. **Pre-Save vs Post-Load `W.byId` Divergence (`N-08 / B-15` Root Cause!)**:
   - On line `488`, splicing a buried NPC from `W.npcs` **leaves `W.byId[n.id]` populated in live RAM**, whereas `SIM.deserialize` (`line 608`) rebuilds `W.byId` **solely from `W.npcs`** (`so W.byId[n.id] becomes undefined after reload!`)!
   - Also clean `PEAKS` (`masterId`/`deputyId`/`memberIds`), `LEYLINE_GRID` (`guardNpcId`), and `OUTPOSTS` (`garrisonNpcIds`) on `AI.kill` / burial, and `delete W.byId[n.id]` when burying into `W.graveyard` (using `W.sect.historicalPeople[id]` for historical name lookups) so live RAM and reloaded state match 100%!
3. **Missing `.ensure(W)` Calls in `SIM.deserialize` (`lines 630–655` vs `256–278`)**:
   - Add the 8 `.ensure(W)` calls present in `SIM.create` (`TOTEMS_BLOODLINES`, `CELESTIAL_ALIGNMENTS`, `DESTINY_KARMA`, `LEYLINE_GRID`, `TRIBUNAL`, `AUTOMATION`, `ensureDefectorPursuit`, `ANCESTRAL_PAVILION`, `ensureJurisdictionLog`) into `SIM.deserialize`!

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3. [SECT-SYS-03] Macro Fiscal Closed-Loop Gaps (Emergency Grain, Monthly Tribute, Facility Upkeep) & mentor vs mentorId Split!

1. **Unlinked Macro Stone Faucets & Sinks (sim.js:518 & sect.js:SECT.monthly)**:
   - Emergency grain buying (sim.js:518: ECO.pay(W, c, 'ซื้อข้าวฉุกเฉิน')), monthly regional protection tribute (SECT.monthly: 150 + 26 * sqrt(fame)), and monthly facility upkeep (ค่าบำรุงรักษาสำนัก — **preserve exact formula tested in v214_governance.js:127!**) currently mutate W.sect.stones without balancing against W.world.resourcePool.stones! Credit/debit W.world.resourcePool.stones whenever W.world && W.world.resourcePool exists so macro world stone supply remains conserved across 100-year runs.
2. **Two Disconnected Mentorship Fields (n.mentor in sim.js:244 vs n.mentorId / n.apprenticeIds in ai.js:887)**:
   - Founding disciples in SIM.create (sim.js:244) get n.mentor = master.id and master.h.life.family.disciples.push(n.id), **without** setting n.mentorId = master.id or master.apprenticeIds = [...]! When the founding Master dies in AI.kill (ai.js:887), master.apprenticeIds is empty—so the founding disciples **never inherit their Master's final mentorImprint Qi legacy or BLACK_THREAD vengeance bond**! Synchronize mentor $\leftrightarrow$ mentorId and family.disciples $\leftrightarrow$ apprenticeIds via a single helper SECT.bindMentorship(W, masterNpc, discipleNpc).

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4. [SECT-SYS-04] Reconciling SECT.AUTOMATION (PROP-35) With True Character-Driven Autonomy

- Currently, SECT.AUTOMATION.runDailyAutomation(W) (v330 Phase 7) acts like an omniscient robot executing tasks even when offices are vacant or corrupt.
- **Architectural Synthesis**: Keep SECT.AUTOMATION APIs 100% compatible with v330, while making **living Officers** the actors in simulation:
  1. autoAssignNewDisciplesToPeaks: Executed by **registry / disciples** officer (if corrupt/nepotistic, favors high-bribe or same-clique Peaks!).
  2. autoRepairLeylines: Executed by **arraymaster / steward** (spending stones -> W.world.resourcePool.stones).

3. `autoDistributeBreakthroughPills`: Executed by **`treasury`** / **`master`**, weighted by `policy.meritocracy` vs `SECT.relationScore` (so nepotistic treasurers spark real `unfair_allocation` petitions!).
4. `autoTriggerClosedDoorOnBottleneck`: Driven **directly by each bottlenecked NPC's own `BN.decide` psychology** (`high caution/discipline chooses enterClosedDoorSeclusion and asks their mentor/companion/sworn-sibling to stand guard as protectorId`!)—defaulting `autoTriggerClosedDoorOnBottleneck` to organic NPC choice without needing the player to flip a toggle!

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5. Master Prioritized Execution Roadmap for GPT & Gemini (`P0` $\to$ `P3` Across `07A–07E`)

Phase	**Priority**	**Target Files**	**Exact Deliverables Across `07A–07E`**
Phase 1	**P0 — Critical State Corruption, Wallet Black-Holes & Broken Jail**	`sim.js`, `sect.js`, `ai.js`	1) Sync `officer.personalStones` $\leftrightarrow$ `officer.inv.stones` in `payOfficeStipends` & schedule monthly (`07A`). 2) Set `offender.jail = W.t + days * 1440` on `SEVERE_DUNGEON_DETENTION` so detained criminals are actually confined (`07D`). 3) Fix `SIM.retire` & `graveyard` (`line 488/583`): preserve `n.inv.stones`/`pets` conservation, stop

Phase	Priority	Target Files	Exact Deliverables Across `07A–07E`
			traitors donating secret scrolls, clean dangling IDs from `PEAKS`/`LEYLINE_GRID`/`OUTPOSTS`, delete `W.byId[id]` on burial, and fix `g2.id` string check + missing `.ensure(W)` calls in `SIM.deserialize` (`07E`).
Phase 2	**P1 — Empty Peak Stations, Homeless Dorms, Totem Sync & Dungeon Bugs**	`ai.js`, `world_map.js`, `sect.js`	1) Route Peak members' `meditate`, `practice`/`dummy`/`lift`, and `bed` requests in `AI.freeStation` & `assignBed` to their founded Peak Sanctuary (`144` stations active!) (`07B`). 2) Sync `SECT.housingCapacity` with physical `'bed'` stations & relocate promoted elders to `dorm_core`/`elder_quarters` (`07C`). 3) Sync `W.sect.totems` onto newly recruited disciples (`07C`), unify `pObj.treasuryStones` $\leftrightarrow$

Phase	Priority	Target Files	Exact Deliverables Across 07A–07E
			pObj.vaults.stones (07B), and fix RECRUIT_DEFECTOR (N-01) & INTERROGATE (N-09) (07D).
Phase 3	**P2 — Closed-Loop Governance & Ghost Keys/Halls**	sect.js, sim.js	1) Enforce Closed-Loop W.world.resourcePool.stones / offender.inv.stones / homeSect.treasury across quartermaster, marketmaster, auditOfficerConduct, launchPeakExpedition, RANSOM, enactSectCharter, and macro upkeep/tribute (07A/07B/07D/07E). 2) Map the 9 broken POST_CATALOG.hall IDs (herb_garden, beast_menagerie, etc.), fix sk.LIGHTFOOT/HERB_LORE + 16-weapon scoring (07A), fix 'P:BASIC_HEAL_1' $\to$

Phase	Priority	Target Files	Exact Deliverables Across `07A–07E`
			`'P:BASIC_HP'` in `OUTPOSTS_LOGISTICS` (`07D`), and normalize `5`-vs-`10` elements on `ELEMENTAL_BARRIER` (`07C`).
Phase 4	**P3 — Full Emergent Autonomy Across All Political & Dynasty Systems**	`sect.js`, `ai.js`, `cognition.js`	1) Auto-unlock specialized bureaus by sect growth; connect `W.sect.petitions` $\to$ `censor` $\to$ `TRIBUNAL` (`07A`). 2) Enable organic `spawnCompanionOffspring` (`3-Gen Clan Dynasty`), Master-driven `CHARTER_DOCTRINE`, seasonal `STRATEGIC_COUNCIL`, real `SEAL_DUEL` power bouts, real `SPLINTER` rebel departure, Peak-Lord building/policy/array autonomy, `arraymaster` war/peace formation switching, `COSMIC_SHRINE` offerings, `INFERNAL_SHRINE` orthodox friction, and unified

Phase	Priority	Target Files	Exact Deliverables Across `07A–07E`
			`bindMentorship` (`07B–07E`).